

Carbon Saving Calculation

Carbon Saving Calculation — Method Explained

YOUR CODE — Line 416:

```
co2_step = p_pv × dt × 0.72  
(kW) (0.25h) (kg/kWh)
```

What each part means

p_pv (kW)

PV power generated
at this 15-min step

Max = 2089.35 kW
at solar noon

This is solar energy
replacing grid energy

dt = 0.25 hours

Time step duration
15 minutes = 0.25h

$p_{pv} \times dt = \text{kWh}$
Energy generated
in this 15-min step

0.72 kg/kWh

CEB Sri Lanka grid
emission factor

= 0.72 kg CO₂
emitted per kWh
of grid electricity

The logic — what "carbon saving" means

Every 1 kWh generated by solar PV = 1 kWh NOT drawn from the CEB grid

Every 1 kWh NOT drawn from grid = 0.72 kg CO₂ NOT emitted by CEB power stations

CO₂ saved = PV energy generated x grid emission factor

This assumes: if PV did not exist, ALL energy would come from the grid

Example at 12:00 (step 48)

$p_{pv} = 1821.60 \text{ kW}$ (peak solar, $1821.60 \text{ kWp} \times 1.0 \text{ irradiance}$)

$dt = 0.25 \text{ h}$

$co2_step = 1821.60 \times 0.25 \times 0.72 = 327.9 \text{ kg CO}_2 \text{ saved in 15 minutes}$
= equivalent to taking 67 cars off the road for one day

Adapted from [this](#) by the same author

Source: Ceylon Electricity Board (CEB) Annual Report / IPCC emissions data

Sri Lanka grid mix 2023: coal 40%, oil 25%, hydro 25%, other renewables 10%

Weighted emission factor = 0.72 kg CO₂ per kWh (published by CEB, 2023)

For your report cite: CEB Annual Report 2023, Grid Emission Factor for Sri Lanka

3 Types of Grid Emission Factor — Sri Lanka 2022

1. Simple Operating Margin (SOM) = 0.7123 kg CO₂/kWh

What it means:

How much CO₂ the grid CURRENTLY emits per kWh from existing running power stations
Calculated from: coal plants + oil plants + gas plants already operating today
Does NOT include hydro and renewables (they have zero CO₂ but run regardless)

When to use: when your solar PV DISPLACES existing fossil fuel generation

Your case: your PV reduces how much coal/oil the CEB burns → use this value

2. Build Margin (BM) = 0.5841 kg CO₂/kWh

What it means:

How much CO₂ the NEXT power plant to be built would emit per kWh
Based on: the last 5 power plants built + planned future capacity additions
Sri Lanka is building more renewables → build margin is lower than operating margin

When to use: when your PV AVOIDS the need to build a new power plant

Less relevant for your project — your PV displaces current generation not future

3. Combined Margin (CM) = 0.6482 kg CO₂/kWh

What it means:

50% Simple Operating Margin + 50% Build Margin (UNFCCC standard formula)
 $CM = 0.5 \times SOM + 0.5 \times BM = 0.5 \times 0.7123 + 0.5 \times 0.5841 = 0.6482$
Accounts for BOTH current displacement AND avoiding future capacity

When to use: UNFCCC CDM projects — carbon credits and formal accounting

Used when a project registers for carbon credits under Clean Development Mechanism

Which one to use for YOUR project

Your simulation uses 0.72 kg/kWh → closest to Simple Operating Margin (0.7123)

This is correct because: your PV displaces currently running fossil fuel generation

When your solar generates at noon → CEB reduces coal/oil plant output → CO₂ saved

Recommend updating code: SRI_LANKA_CO2 = 0.7123 (more accurate official value)

SOM = 0.7123 → What existing dirty plants emit → USE THIS for displacement projects

BM = 0.5841 → What new plants would emit → Use for capacity planning projects

CM = 0.6482 → Average of both (UNFCCC standard) → Use for carbon credit applications

Your project → Simple Operating Margin → 0.7123 kg CO₂/kWh is the right choice

<https://www.energy.gov.lk/images/energy-balance/energy-balance-2022.pdf>

9.1.1 Operating Margin

The Operating Margin (OM) is a concept which includes all power plants which can have reduced outputs due to a project. It specifically excludes 'low cost, must run' power plants, implying that with or without the project, such generation will continue. Table 9.3 gives the Simple Operating Margin (OM).

Table 9.3 – Operating Margin

	2019	2020	2021	2022
Emissions from Power Plants (t-CO ₂)	3,552,816.2	2,960,912	1,908,745	1,642,648
Net Electricity Generation (GWh) excluding low-cost must run power plants	5,006.7	4,179.3	2,498.4	2,464.5
Operating margin CO ₂ emission factor (kg-CO ₂ /kWh)				
Three-year generation based weighted average	0.7084	0.7084	0.7208	0.7123

9.1.2 Build Margin

The Build Margin (BM) is a concept which attempts to foretell the happenings of a generation system in future, during the crediting period of a project, considering the recent additions to a generation system.

Table 9.4 – Build Margin

	Unit	2019	2020	2021	2022
Emissions of power plants considered for the BM	tonnes of CO ₂	4,266,621.5	4,111,940.3	3,922,524.3	3,290,087.8
Generation of power plants considered for the BM	GWh	5,101.3	5,381.3	5,863.0	5,632.3
Build margin emission factor	kg-CO ₂ /kWh	0.8364	0.7641	0.6690	0.5841

9.1.3 Combined Margin

The Combined Margin (CM) is a weighted average of OM and BM and is commonly known as the Grid Emission Factor (Table 9.5).

Table 9.5 – Combined Margin (kg-CO₂/kWh)

	2019	2020	2021	2022
For solar, wind Projects	0.7404	0.7224	0.7079	0.6803
All other Projects; 1 st crediting period	0.7724	0.7363	0.6949	0.6482
All other Projects; 2 nd - 3 rd crediting period	0.8044	0.7502	0.6820	0.6162

The OM, BM and CM are required for the assessment of CO₂ emission reductions for projects claiming carbon credits under UNFCC guidelines. The GEF is indicated in Figure 9.1.